# Bayes' Updating

## Video Lesson

Duration: 00:12:47

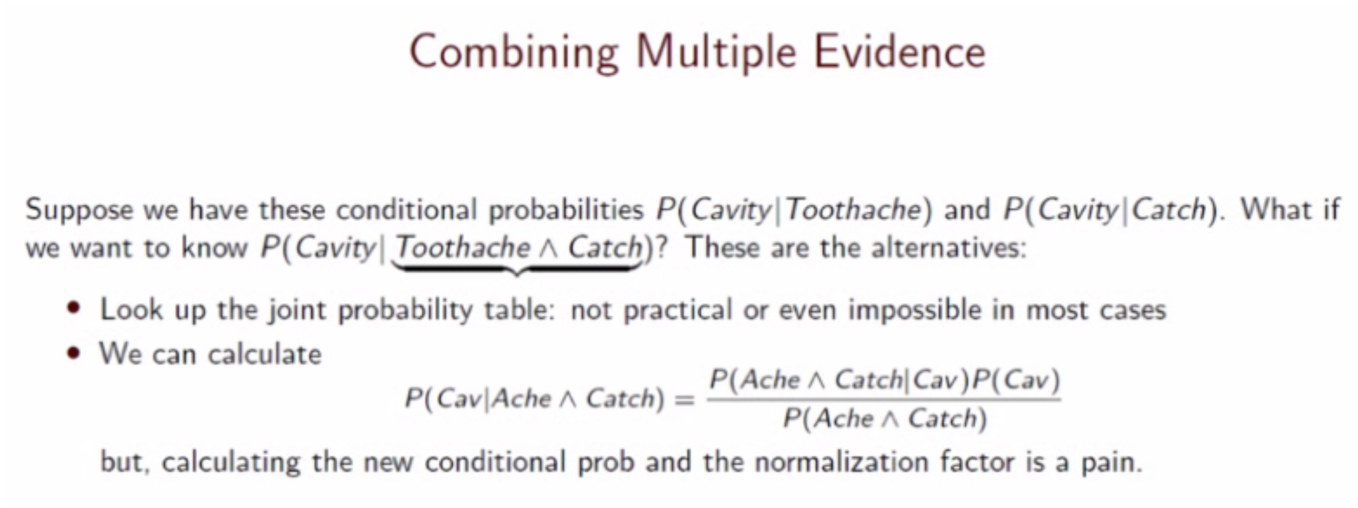
## Slide Deck

**M11S06 (https://canvas.tamu.edu/courses/430127/files/92570692?wrap=1)**

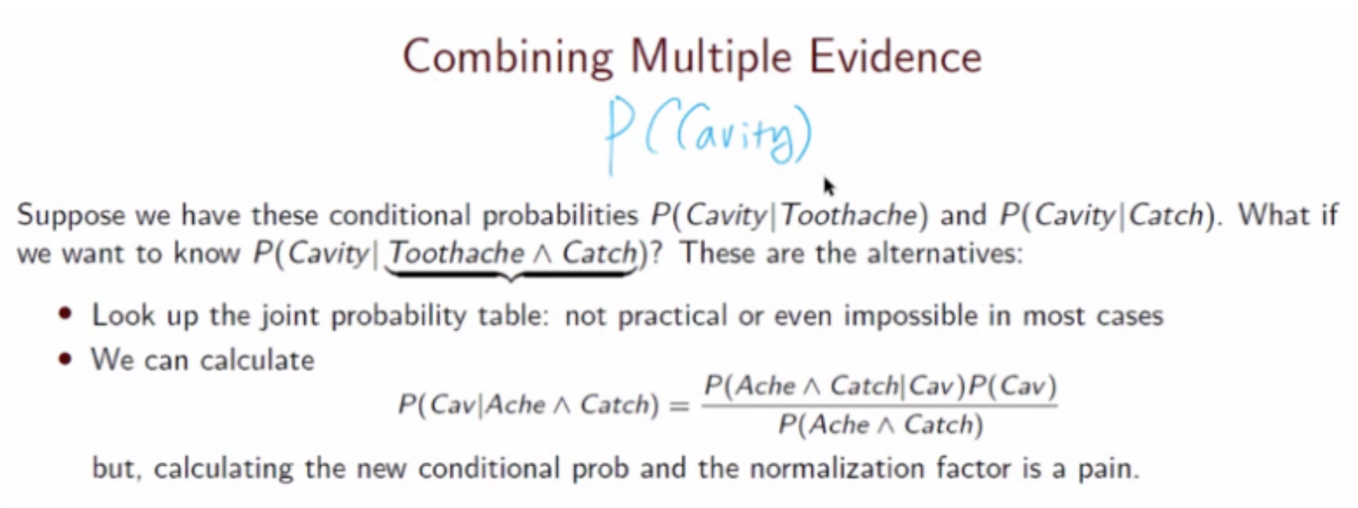
## Lesson Transcript

In this lecture, we will look at Bayesian Updating. Bayesian updating makes incorporating multiple pieces of evidence more efficient.

---




Let us consider a situation where we must combine multiple pieces of evidence.




Let us first consider this probability. What is the probability of we having a cavity? Here we do not have any further information, so it is just a wild guess. But what if we receive two pieces of evidence, toothache and cavity?

If we treat these two individually, then we have this conditional probability. Probability of cavity given toothache and probability of cavity given catch.

## Combining Multiple Evidence

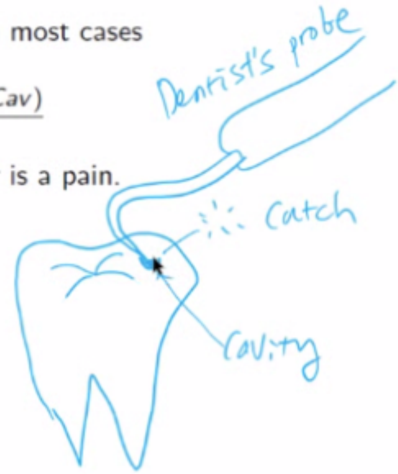
$P(Cavity)$

Suppose we have these conditional probabilities $P(Cavity|Toothache)$ and $P(Cavity|Catch)$. What if we want to know $P(Cavity|\underbrace{Toothache \wedge Catch})$? These are the alternatives:

- Look up the joint probability table: not practical or even impossible in most cases
- We can calculate

$$P(Cav|Ache \wedge Catch) = \frac{P(Ache \wedge Catch|Cav)P(Cav)}{P(Ache \wedge Catch)}$$

but, calculating the new conditional prob and the normalization factor is a pain.




We know what toothache is. What is a catch? The catch is described below. So, we have a tooth with a cavity, and you went to the dentist, and the dentist used his or her probe to scratch across the surface of your tooth. And when this tip actually goes inside the cavity, it will catch instead of just sliding over.

And that is called the catch in technical terms.

And of course, as you can see, toothache and catch both indicate that you have a cavity. Maybe this is a kind of hollow dent in your tooth. Maybe just due to some kind of a breakage instead of a cavity. So, everything is not exactly determined.

---

## Combining Multiple Evidence

Suppose we have these conditional probabilities $P(Cavity|Toothache)$ and $P(Cavity|Catch)$. What if we want to know $P(Cavity|\underbrace{Toothache \wedge Catch})$? These are the alternatives:

- Look up the joint probability table: not practical or even impossible in most cases
- We can calculate

$$P(Cav|Ache \wedge Catch) = \frac{P(Ache \wedge Catch|Cav)P(Cav)}{P(Ache \wedge Catch)}$$

but, calculating the new conditional prob and the normalization factor is a pain.

The main question that we want to answer here is, instead of dealing with these two separately, what if you combine these two? Then what would your belief about the cavity be? Would it change? Would it be the same? You can ask these interesting questions.

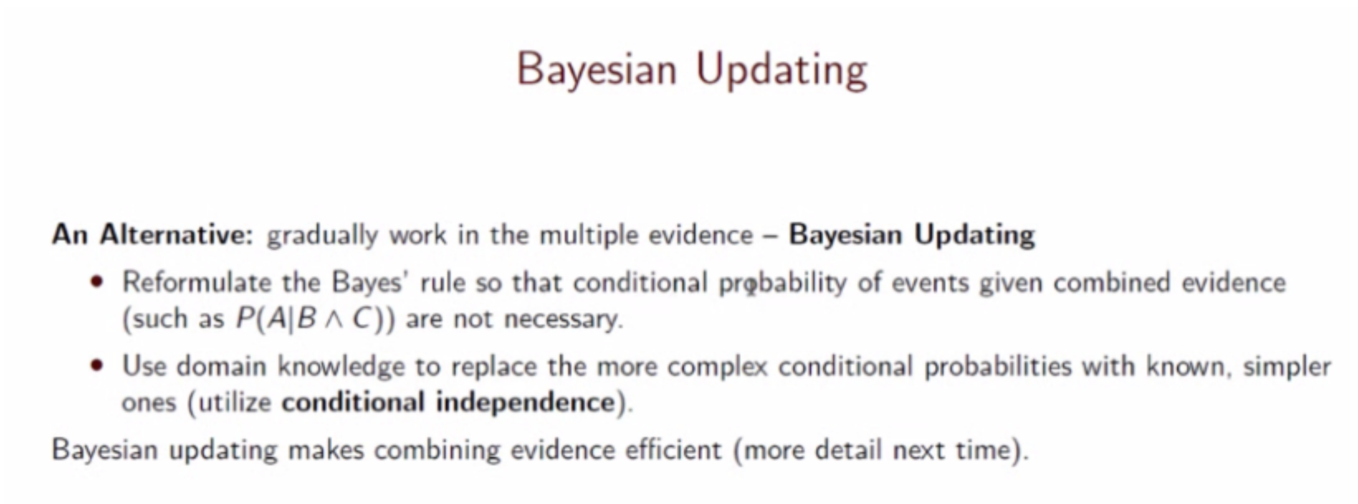
In the general case, you can keep on adding new pieces of evidence like toothache, catch, bad breath, and so on, and generally, your belief about cavity will change as you keep on piling more and more pieces of evidence.

That is all fine, but how can you compute this posterior probability given more than one piece of evidence? Since we learned about the Bayes' Rule, our first attempt is just to use Bayes' Rule.

So, cavity, given ache and catch, would become ache and catch given cavity multiplied by the prior probability of cavity, divided by the probability of ache and catch.

But it looks like these are also quite difficult to compute. Of course, you can always look up the joint probability table, but this is not practical, as we saw last time.

---




Bayesian Updating is the best method that you can use to solve this kind of issue. So, the idea is to gradually work on the multiple pieces of evidence using a technique called Bayesian Updating.

In this approach, we want to reformulate the Bayes' Rule so that conditional probability of events given combined evidence such as *P* of *A* given *B* and *C* is unnecessary.

Here domain knowledge comes in very handy. We want to use domain knowledge to replace the more complex conditional probabilities with known and simpler ones.

Here, the property that we use is called conditional independence.

Following this approach, Bayesian Updating makes combining evidence very efficient.

---

## Bayesian Updating: Example

Goal: only use

$p(Cav)$

$p(Ache|Cav)$

$p(Catch|Cav)$

We want to calculate $P(Cavity|Ache \wedge Catch)$:

$$P(Cav|Ache \wedge Catch) = P(Cav|Ache)\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}$$

$$= \underbrace{P(Cav)\frac{P(Ache|Cav)}{P(Ache)}}_{old}\underbrace{\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}}_{new}$$

Problem is that $P(Catch|Ache \wedge Cav)$ may be equally hard to calculate. However, we can make these assumptions (**Conditional Independence** of *Ache* and *Catch* given *Cav*):

$$\begin{aligned} P(Catch|Cav \wedge Ache) &= P(Catch|Cav) \\ P(Ache|Cav \wedge Catch) &= P(Ache|Cav) \end{aligned}$$

Only thing that remains is $P(Ache)P(Catch|Ache) = P(Catch \wedge Ache)$, which can be eliminated by normalization.

Hint: Use Extended Bayes Rule: $P(Cav|Catch, Ache) = \frac{P(Catch|Cav, Ache)P(Cav|Ache)}{P(Catch|Ache)}$.

Thus, again, consider the probability of cavity given toothache and catch. The goal here is to use only these three pretty easy-to-obtain probabilities. The prior probability, the probability of cavity, and these two likelihood probabilities, probability of toothache given cavity, probability of catching given cavity.

Through some manipulations, you can find that this probability of a cavity, given toothache and catch becomes this formula here. Here we find that there is the probability of a cavity, the first one, the probability of toothache given cavity, the second one, and something sort of close to the third one, the probability of catching, given toothache, but there is this extra cavity that is added on the right-hand side of this bar.

Assuming that the normalization factors are the same, we can see that these three would be enough only if we can make this to be the same as that one.

## Bayesian Updating: Example

Goal:
only use
$p(Cav)$
$p(Ache|Cav)$
$p(Catch|Cav)$

We want to calculate $P(Cavity|Ache \wedge Catch)$:

$$
\begin{aligned}
P(Cav|Ache \wedge Catch) &= P(Cav|Ache)\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)} \\
&= \underbrace{P(Cav)\frac{P(Ache|Cav)}{P(Ache)}}_{old}\ \underbrace{\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}}_{new}
\end{aligned}
$$

Problem is that $P(Catch|Ache \wedge Cav)$ may be equally hard to calculate. However, we can make these assumptions (**Conditional Independence** of $Ache$ and $Catch$ given $Cav$):

$$
\begin{aligned}
P(Catch|Cav \wedge Ache) &= P(Catch|Cav) \\
P(Ache|Cav \wedge Catch) &= P(Ache|Cav)
\end{aligned}
$$

Only thing that remains is $P(Ache)P(Catch|Ache) = P(Catch \wedge Ache)$, which can be eliminated by normalization.

Hint: Use Extended Bayes Rule: $P(Cav|Catch, \cancel{Ache}) = \frac{P(Catch|Cav, \cancel{Ache})P(Cav|\cancel{Ache})}{P(Catch|\cancel{Ache})}$.

For this very first step, we will simply use the Extended Bayes' Rule that we saw earlier. The full formula is the probability of cavity given catch, and toothache equals the probability of catch given cavity and toothache, multiplied by the probability of cavity given toothache, divided by the probability of catch given toothache.

Again, one way of looking at the Extended Bayes' Rule is if you ignore this ache part that is common in all four of these probabilities, then it is simply the same as the Bayes' Rule. So, just look at the green-marked probabilities. The only change is now you added this extra conditioning random variable toothache to the right-hand side.

Like that, like that, like that, like that. So, that is how we use this Extended Bayes' Rule.

## Bayesian Updating: Example

We want to calculate $P(Cavity|Ache \wedge Catch)$:

$$P(Cav|Ache \wedge Catch) = P(Cav|Ache)\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}$$
$$= \underbrace{P(Cav)\frac{P(Ache|Cav)}{P(Ache)}}_{old}\underbrace{\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}}_{new}$$

Problem is that $P(Catch|Ache \wedge Cav)$ may be equally hard to calculate. However, we can make these assumptions (**Conditional Independence** of *Ache* and *Catch* given *Cav*):

$$\begin{aligned} P(Catch|Cav \wedge Ache) &= P(Catch|Cav) \\ P(Ache|Cav \wedge Catch) &= P(Ache|Cav) \end{aligned}$$

Only thing that remains is $P(Ache)P(Catch|Ache) = P(Catch \wedge Ache)$, which can be eliminated by normalization.

Hint: Use Extended Bayes Rule: $P(Cav|Catch, Ache) = \frac{P(Catch|Cav, Ache)P(Cav|Ache)}{P(Catch|Ache)}$.

We can compare that with this formula here. There is the probability of a cavity given toothache, which is this one here. The probability of catching, given toothache and cavity is this one, the likelihood, divided by the normalization factor probability of catching, given toothache, which is that.

Again, this step just required using the Extended Bayes' Rule.

Then the next step is just to carry this over and apply the Bayes' Rule on this again. So, the probability of a cavity, given toothache, is the probability of toothache given cavity. Do the switch, and now we have the probability of cavity as the prior, divided by the normalization factor. Again, from here to here, you just use the same Bayes' Rule.

## Bayesian Updating: Example

We want to calculate $P(Cavity|Ache \wedge Catch)$:

$$P(Cav|Ache \wedge Catch) = P(Cav|Ache)\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}$$

$$= \underbrace{P(Cav)\frac{P(Ache|Cav)}{\cancel{P(Ache)}}}_{old}\underbrace{\frac{P(Catch|Ache \wedge Cav)}{\cancel{P(Catch|Ache)}}}_{new} = P(Catch|Cav)?$$

Problem is that $P(Catch|Ache \wedge Cav)$ may be equally hard to calculate. However, we can make these assumptions (**Conditional Independence** of $Ache$ and $Catch$ given $Cav$):

$$P(Catch|Cav \wedge Ache) = P(Catch|Cav)$$
$$P(Ache|Cav \wedge Catch) = P(Ache|Cav)$$

Only thing that remains is $P(Ache)P(Catch|Ache) = P(Catch \wedge Ache)$, which can be eliminated by normalization.

Hint: Use Extended Bayes Rule: $P(Cav|Catch, Ache) = \frac{P(Catch|Cav, Ache)P(Cav|Ache)}{P(Catch|Ache)}$.

Again, I am assuming that whether you have a cavity or not, this denominator part would be the same. Let us say we ignore this for now. Then you only need to find out whether this can be reduced to that, that is, the probability of catching, given toothache and cavity equals the probability of catching, given cavity. And that is exactly written here.

And in this situation, we can use something called conditional independence to make this equality to be true.

## Bayesian Updating: Example


We want to calculate $P(Cavity|Ache \wedge Catch)$:

$$P(Cav|Ache \wedge Catch) = P(Cav|Ache)\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}$$
$$= \underbrace{P(Cav)\frac{P(Ache|Cav)}{P(Ache)}}_{old}\ \underbrace{\frac{P(Catch|Ache \wedge Cav)}{P(Catch|Ache)}}_{new}$$

Problem is that $P(Catch|Ache \wedge Cav)$ may be equally hard to calculate. However, we can make these assumptions (**Conditional Independence** of $Ache$ and $Catch$ given $Cav$):

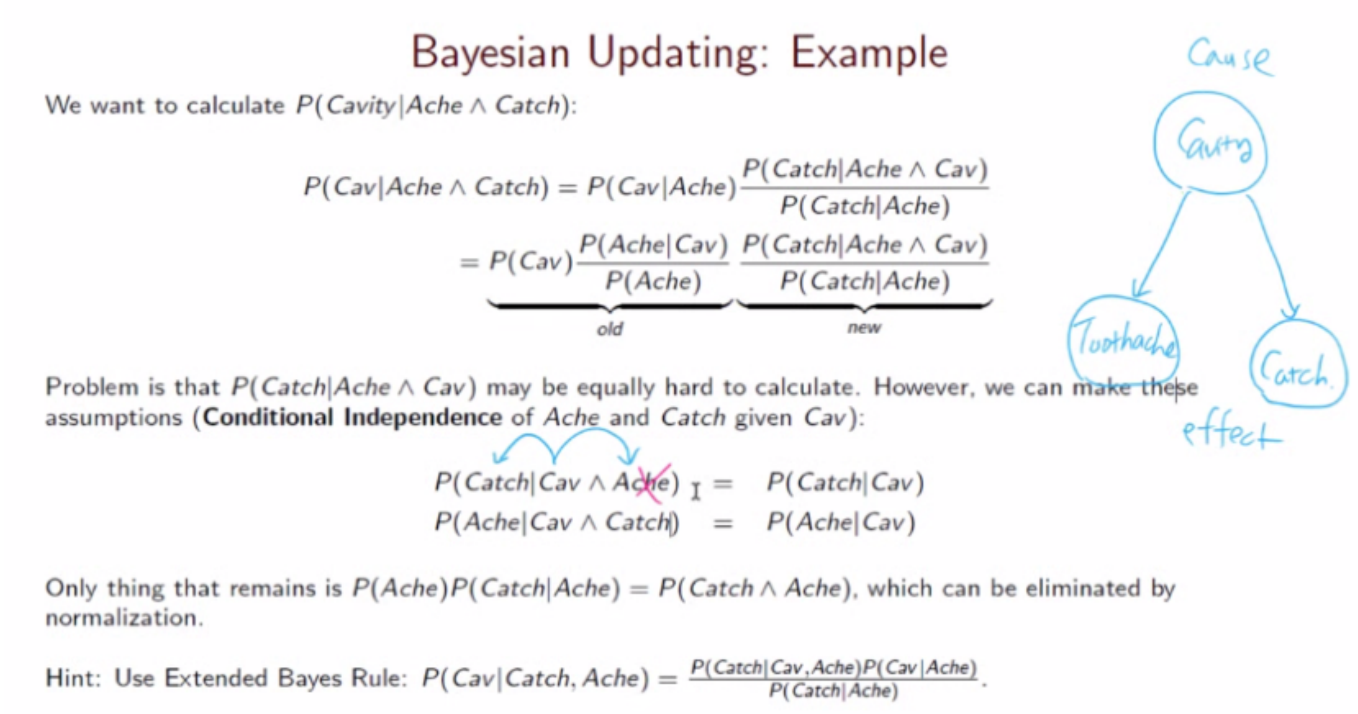
$$\begin{aligned} P(Catch|Cav \wedge Ache) &= P(Catch|Cav) \\ P(Ache|Cav \wedge Catch) &= P(Ache|Cav) \end{aligned}$$

Only thing that remains is $P(Ache)P(Catch|Ache) = P(Catch \wedge Ache)$, which can be eliminated by normalization.

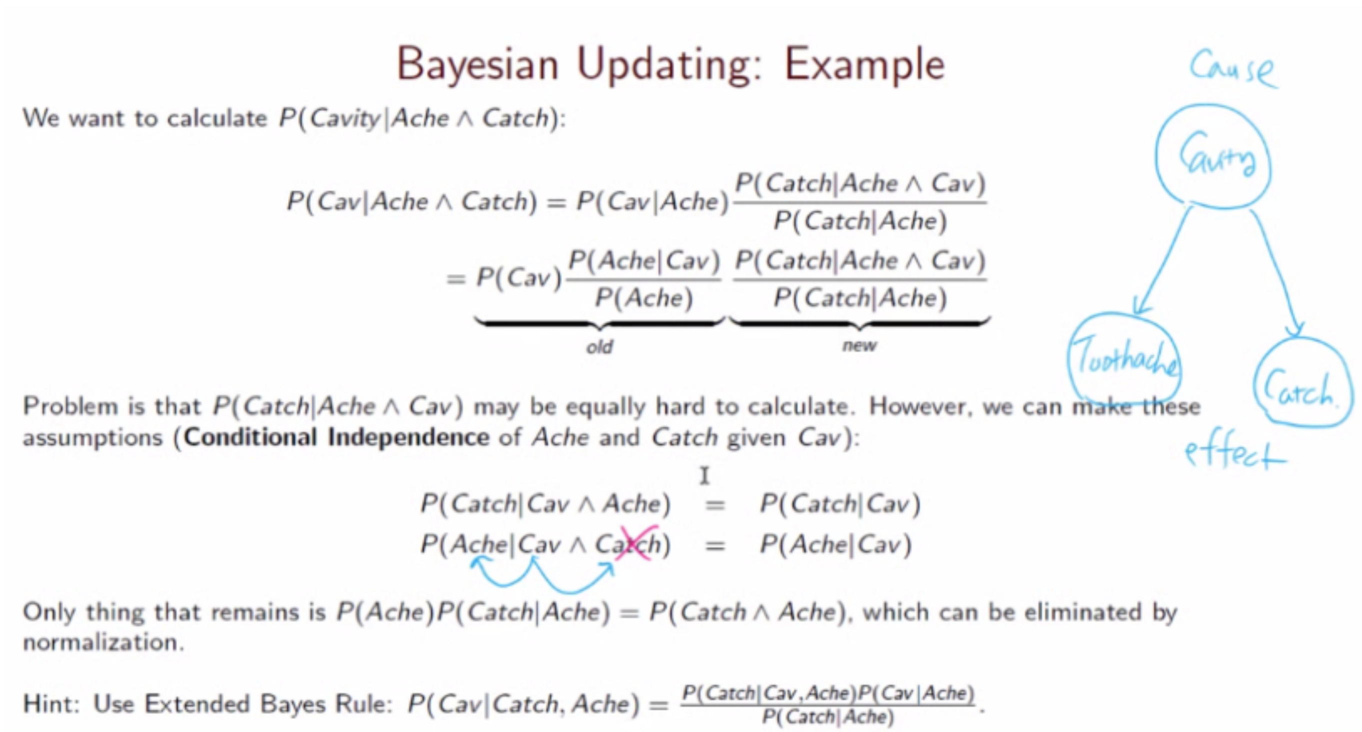
Hint: Use Extended Bayes Rule: $P(Cav|Catch, Ache) = \frac{P(Catch|Cav, Ache)P(Cav|Ache)}{P(Catch|Ache)}$.

We can reason about this equality by first looking at this relationship. Here is cavity, and it is the cause of both toothache and catch. In other words, toothache and catch are effects of the common cause of cavity. And when you project that relationship to this conditional probability, you can see that this cavity, which is a common cause, would result in two effects, catch, and toothache.

The idea here is that cavity is the cause of catch; once you know that you have a cavity, then whether there will be a catch or not will be quite evident without any extra information.




This toothache information is not needed. And even if you have that, it does not change your estimate of the probability of catch.




This is the same for the second example. The probability of toothache given cavity and catch equals the probability of toothache given cavity. Again, cavity is a common cause of toothache and catch. So, once you learn about cavity, then catch becomes redundant information. Thus, you can eliminate it without changing your belief about toothache.

## Bayesian Updating: Example (cont'd)

So, after replacing the factors:

$$P(Cav|Ache \wedge Catch) = \alpha \underbrace{P(Cav)P(Ache|Cav)}_{old} \underbrace{P(Catch|Cav)}_{new},$$

where $\alpha$ is the normalization constant needed to make $P(Cav|Ache \wedge Catch)$ add up to 1.

To summarize, Bayesian Updating. Again, you start with the probability of cavity, given toothache and catch. Here again, we have two pieces of evidence. How can you compute this posterior probability without going through multiple levels of the complex derivation of different conditional probabilities? Simply you just multiply the prior probability of cavity, multiplied by the likelihood, probability of toothache given cavity multiplied by the probability of catch given cavity.

## Bayesian Updating: Example (cont'd)

So, after replacing the factors:

$\wedge$ Bad Breath

$$P(Cav|Ache \wedge Catch) = \alpha \underbrace{P(Cav)P(Ache|Cav)}_{old} \underbrace{P(Catch|Cav)}_{new}, \; P(Bad\ Breath|Cav)$$

where $\alpha$ is the normalization constant needed to make $P(Cav|Ache \wedge Catch)$ add up to 1.

On top of toothache and catch, let us say you received another piece of evidence, bad breath, then updating this posterior probability is as simple as just multiplying the probability of bad breath given cavity, and you will adjust this α.

## Bayesian Updating: Example (cont'd)

So, after replacing the factors:

$$P(Cav|Ache \wedge Catch) = \alpha\, \underbrace{P(Cav)P(Ache|Cav)}_{old}\, \underbrace{P(Catch|Cav)}_{new},$$

where $\alpha$ is the normalization constant needed to make $P(Cav|Ache \wedge Catch)$ add up to 1.

$$P(Cav|Ache \wedge Catch) = \alpha \cdot p(Cav)\, P(Ache|Cav)$$

$$\parallel$$

$$P(\neg Cav|Ache \wedge Catch) = \alpha \cdot p(\neg Cav) \cdot p(Ache|Cav)$$

So again, let me explain a little bit about this α, the normalization constant. Let us consider these two posterior probabilities, the probability of cavity, given toothache and catch, and the probability of not cavity, given toothache and catch. Because these two are inclusive, if you add them, it must result in one. And because this normalization factor α only contains probabilities about toothache and catch, they are the same regardless of this cavity, whether it is true or false.

So, if you just compute this and compute that, then without this, they will add up to some value that is greater than 1. Then you just divide this by the sum of these two, then you will get the correct value. Again, this α would be basically the sum of this and that divided by 1.

## Bayesian Updating: Example (cont'd)

So, after replacing the factors:

$$P(Cav|Ache \wedge Catch) = \alpha \underbrace{P(Cav)P(Ache|Cav)}_{old} \underbrace{P(Catch|Cav)}_{new},$$

where $\alpha$ is the normalization constant needed to make $P(Cav|Ache \wedge Catch)$ add up to 1.

$$P(Cav|Ache \wedge Catch) = \alpha \cdot p(Cav)\,p(Ache|Cav)$$

$$\parallel$$

$$p(\neg Cav|Ache \wedge Catch) = \alpha \cdot p(\neg Cav) \cdot p(Ache|Cav)$$

$$\alpha = \frac{1}{p(Cav)p(Ache|Cav) + p(\neg Cav)p(Ache|Cav)}$$

To repeat, α can be computed from these two as 1 over the first term here plus the second term here.

---

## Bayesian Updating: Summary

$X$ and $Y$ are independent given $Z$ (conditional independence):

$$P(X|Y, Z) = P(X|Z)$$

Simplified Bayes' rule for multiple evidence is[1]:

$$P(Z|X, Y) = \alpha P(Z)P(X|Z)P(Y|Z),$$

where $\alpha$ is the normalization constant.
Thus, Bayesian Updating makes combining multiple evidence easy.

In summary, in Bayesian Updating, conditional independence plays a key role. X and Y are independent given Z, which is expressed as *X* given *Y,* and *Z* equals the probability of *X* given *Z*.

Again, for this to be valid, *Z* must be the common cause of both *X* and *Y*.

---

## Bayesian Updating: Summary

$X$ and $Y$ are independent given $Z$ (conditional independence):

$$P(X|Y,Z) = P(X|Z)$$

Simplified Bayes' rule for multiple evidence is[1]:

$$P(Z|X,Y) = \alpha P(Z)P(X|Z)P(Y|Z),$$

where $\alpha$ is the normalization constant.
Thus, Bayesian Updating makes combining multiple evidence easy.

From this, we derive the Bayes' Rule for multiple pieces of evidence as the probability of *Z* given *X*, *Y* equals α, the normalization factor, times the prior probability *P* of *Z* times *P* of *X* given *Z*. That is the first piece of evidence. And *P* of *Y* given *Z*. That is the second piece of evidence.

Using this Bayesian Updating makes combining multiple pieces of evidence easy.